Chapter 4

Modeling Causal Impact via Randomized Experiments

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Just as the rapid adoption of the telephone sparked Erlang's stochastic models for network congestion at the turn of the 20th century, the severe food shortages in 1920s Europe following World War I pushed statisticians to develop better methods for evaluating crop varieties and soil conditions to improve agricultural yields.

Enter the protagonist of this chapter, Jerzy Neyman, who from 1917 to 1921 served as a statistician at the Institute of Agriculture in Bydgoszcz, Poland. His seminal work in 1923 would lay the foundation for modern causal inference, the scientific and business impact of which can hardly be understated.

From the viewpoint of modeling, Neyman's overarching goal was remarkably similar to Erlang's: to develop an effective model of a complex reality that, when calibrated with observational data, would provide causal guidance on how to achieve better outcomes. However, a closer inspection reveals some striking and fascinating divergences between the two approaches:

1. Whereas Erlangian models are highly mechanistic, we will see that Neyman's causal inference framework does not assume we can (or even need to) accurately describe those underlying mechanisms. Rather than modeling reality as interacting submodules governed by explicit rules, Neyman takes a more end-to-end approach, directly estimating the causal impact of an intervention instead of laboring to map the inner workings of the system in question.

2. Whereas Erlangian models tend to rely on a deterministic core structure (such as the mechanisms governing buffers and servers) to describe a highly stochastic system, the original Neymanian framing is almost the opposite: it aims to estimate *deterministic* treatment effects by intentionally introducing randomness into the assignment process.

We will begin this chapter by walking through a basic version of Neyman's result using more modern notation and terminology developed by Rubin. We will then take a deeper dive into these two divergences and see what we can learn from this new way of thinking about impact and causality.

4.1 Learning Treatment Effects via Randomization

Potential outcomes.

Neyman's original motivation for experimentation remains a great starting point. Suppose we have at our disposal two crop varieties $w \in \{0,1\}$, and we would like to understand which variety will lead to higher yields on average if planted on all lands in the nation. To answer this question, we start with n experimental agricultural plots, with varying soil acidity and other composition factors, which are believed to roughly capture the plot mix in the country at large.

Formally, we call this a randomized control trial with n units (plots), indexed by $i = 1, \ldots, n$. Each unit receives a binary treatment $W_i \in \{0,1\}$ (crop variety) and yields an observed outcome Y_i (crop yield). Conceptually, Neyman posited that the "potential" yields under different treatments $Y_i(1)$ and $Y_i(0)$ are predetermined in theory, but not observed. Thus, in theory, for unit i, we can define this individual treatment effect as

$$\Delta_i = Y_i(1) - Y_i(0),$$

that is, the difference in outcomes between choosing the treatment and choosing the control intervention.

However, in reality, the decision maker can observe its value only once a specific crop has been planted. As such, whatever variety one decides to plant on plot i, one is prevented from seeing the counterfactual outcome of "what would have happened if I had planted the other variety on i." That is, only *one* potential outcome, the one associated with the chosen treatment, can be observed through the experiment. In other words, the individual treatment effect can never be fully observed.[1]

The core insight of Neyman (Neyman, 1923) was that although Δ_i itself is unobservable, randomization allows us to learn averages of these effects. In finite samples, define the sample average treatment effect (SATE):

$$\Delta = \frac{1}{n}\sum_{i=1}^{n}\big(Y_i(1) - Y_i(0)\big).$$

Difference-in-Means Estimation.

A natural estimator for measuring the causal impact of a treatment is the treated-control **d**ifference in sample means:

$$\hat{\tau}_{DM} = \frac{1}{n_1}\sum_{W_i=1} Y_i - \frac{1}{n_0}\sum_{W_i=0} Y_i, \qquad n_w = |\{i : W_i = w\}|.$$

To study this estimator, we impose two further assumptions. First, SUTVA (the Stable Unit Treatment Value Assumption):

$$Y_i = Y_i(W_i), \qquad i = 1, \ldots, n.$$

SUTVA implies two key conditions. First, it means there is no interference between units, so that unit i's outcome depends only on W_i, not on others' assignments. Second, it implies that there are no hidden versions of the treatment, meaning the intervention is uniform enough that $Y_i(1)$ represents a single, unambiguous potential outcome for each unit, rather than varying effects from different doses or versions of the treatment. Together, these rule out spillovers and allow us to cleanly interpret the observed treated-control contrast as a causal contrast between $Y_i(1)$ and $Y_i(0)$.

The second assumption is that the treatment is completely randomized conditional on a fixed number of treated units n_1.

$$\mathbb{P}(W_i = 1 \mid \{Y_i(0), Y_i(1)\}_{i=1}^{n}, n_1) = \frac{n_1}{n}, \qquad i = 1, \ldots, n.$$

These assumptions yield the following foundational result, which shows that the difference in means from the randomized trial gives an unbiased estimate of the *causal* effect:

[1] Note that even trying two different assignments on the same unit sequentially is deemed insufficient, since the nature of the unit may have changed in the meantime.

Theorem 4.1.1. *Under the assumptions above,*

$$\mathbb{E}[\hat{\tau}_{DM} \mid \{Y_i(0), Y_i(1)\}_{i=1}^n,\ n_0 > 0,\ n_1 > 0] = \Delta.$$

The proof is short and relies on linearity of expectation, so we give it here:

Proof. When $n_1 > 0$, we have that

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{n_1}\sum_{W_i=1} Y_i\right] &= \mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^n W_i Y_i \,\middle|\, \{Y_i(0), Y_i(1)\}_{i=1}^n,\ n_1\right] \\
&= \mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^n W_i Y_i(1) \,\middle|\, \{Y_i(0), Y_i(1)\}_{i=1}^n,\ n_1\right] \quad \text{(SUTVA)} \\
&= \frac{1}{n_1}\sum_{i=1}^n Y_i(1)\, \mathbb{E}[W_i \mid \{Y_i(0), Y_i(1)\}_{i=1}^n,\ n_1] \quad (Y_i(1) \text{ is fixed in the conditioning set}) \\
&= \frac{1}{n}\sum_{i=1}^n Y_i(1). \quad \text{(Randomization of treatments assumption)}
\end{aligned}$$

An analogous argument gives

$$\mathbb{E}\left[\frac{1}{n_0}\sum_{W_i=0} Y_i\right] = \frac{1}{n}\sum_{i=1}^n Y_i(0),$$

and subtracting the two equations above yields the desired result. □

4.2 Neymanian versus Erlangian Approaches to Stochastic Modeling

Theorem 4.1.1 may appear deceptively simple, but it is in fact very powerful. So let's unpack a few things.

Foundation for Causality Neymanian randomized trials allow one to directly estimate the causal effect Δ: what would have been the resulting difference in outcome between applying treatment versus control to all units? This causal effect derives its power from the underlying potential outcome assumption: that the outcome given a certain action was already predetermined, and thus not influenced by what the decision maker chooses to implement. The only effect the decision maker has on the world is that of revealing a particular outcome through their action. The causal power the decision maker asserts derives from this fundamental view of the world.

This view towards reality seems very different from that of the Erlangian models. There, the core mechanisms of reality are assumed to be understood (e.g., how queues work). The causal power of the model for informing decisions simply derives from the assumed causal mechanisms themselves.

End-to-End Measurements What is more striking is that Neymanian modeling seems to get to causality without any attempt at elucidating the internal mechanism of the reality in question. To assess the difference in yields from planting different crop varieties, no attempt was made to do so by clarifying the underlying biological pathways or soil chemistries. Instead, the plan was to simply conduct randomized trials and compare the average outcomes in control versus treatment groups. A far cry from the detailed mechanistic models of the Erlangian approach!

Such a seemingly radical approach to modeling, which emerged with Neyman's work, would appear to have stemmed from necessity as much as it did from practicality. Unlike Erlang's model for human-designed queueing systems, Neyman's reality consists of complex biology and soil chemistry, for which understanding the detailed mechanism (if at all possible) would have been a much more daunting task. On the other hand, Neyman's main motivation was intensely practical: to simply understand which crop would have led to higher yields, for which task the scientific understanding of the underlying mechanism may be a luxurious "good to have." Just like an old Chinese adage: 知其然，不知其所以然 (Know that it is, but don't know why it is.)

Remarkably, Neyman himself had done work on both sides of the spectrum and offered his own perspective on the comparative merits of end-to-end versus mechanistic approaches. In a later 1967 paper with Elizabeth Scott, they highlighted that when the main goal was to assess population-level effectiveness, a coarse "somatic" level of modeling suffices, whereas for scientific inquiries where understanding the mechanism is, by definition, the focus, finer-grained mechanistic modeling is more appropriate (Neyman and Scott, 1967). Interestingly, Neyman's arguments there do not seem to concern the modeling of human-designed systems, such as those studied by Erlang.

4.3 Recommended Reading

Historically, the potential-outcomes framework for causal inference traces its origins to Jerzy Neyman's 1923 work on agricultural experiments (See the 1990 English translation of the original, (Neyman, 1923)). Although Neyman and R.A. Fisher were developing foundational experimental methodologies during the same era (with Fisher formulating his randomization-based design methodology (Fisher, 1935)), their early efforts were independent. Both were separately motivated by the practical demands of early 20th-century agricultural research stations: Neyman at the Institute of Agriculture in Bydgoszcz, Poland (Encyclopaedia Britannica, 2026a), and Fisher at the Rothamsted Experimental Station in the UK (Box, 1978). Decades later, Donald Rubin modernized and extended Neyman's framework to accommodate both randomized and nonrandomized observational settings (Rubin, 1974); see also the excellent review in (Holland, 1986). Rubin is also explicitly credited with formalizing the Stable Unit Treatment Value Assumption (SUTVA) (Rubin, 1980). Theorem 4.1.1 and the framing leading up to it is based on Chapter 1 of (Wager, 2024).

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